

CO1 - AT2 Concept Mapping

DBMS software Architecture
Flow from User Request to Data Retrieval



SQL Query/
Request

QUERY

PROCESSOR



Parser

Checks syntax
and semantics of
the query



Query
Optimizer

Generates the
most efficient
execution plan



Query
Executor

Executes the query
plan and requests
data from storage

STORAGE MANAGER



Authorization
Manager

Checks user access
and permissions



Buffer
Manager

Manages data in
memory buffer to
improve access speed



File Manager

Manages files on disk
and space allocation

DATABASE Files (ON DISK)



Data
files

Stores actual
data



Index
files

Stores indexes
for fast search



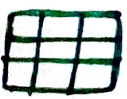
Log files

Stores transaction
logs for recovery



Data Retrieved

Requested data is read from database
files



Query Result

Result is formatted and returned to
the user

ER Diagram Components

1. Entity

Strong Entity

Exists independently and has its own key

Weak Entity

- Depends on strong entity for existence
- Has a partial key (discriminator)
- Connected to strong Entity through identifying relationship.

3. Relationships

One to One

◇ (1:1)

Each entity in A is associated with atmost one entity in B and vice versa.



One to Many

◇ (1:N)

One entity in A can be associated with many in B, but each entity in B is associated with atmost one in A.



Many to Many

◇ (M:N)

Many entities in A can be associated with many entities in B and vice versa.



2. Attributes

Simple Attribute
Cannot be divided furtherComposite Attribute
Can be divided into smaller sub-attributesSingle-valued Attribute
Holds only one value for each entityMulti-valued attribute
Can hold multiple values for each entityDerived Attribute
Value is derived from other attributes.

4. Participation Constraints

Total Participation (Mandatory)

Every entity in the set participates in atleast one relationship



Partial Participation (Optional)

Some entities may or may not participate in the relationship.

